MATH 129: Big Data
HW 7
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Problem 1: Murphy 11.3 - EM for Mixtures of Bernoullis

derive ML update:

We want to prove $\mu_{kj} = \frac{\sum_i r_{ik} x_{ij}}{\sum_i r_{ik}}$

Here,
- $x_{ij} \in \{0,1\}$ is feature j of observation i
- r_{ik} is the responsibility of mixture component k for observation i
- μ_{kj} is the probability that feature $j = 1$ under component k

Bernoulli distribution: $p(x_{ij} | z_i = k) = \mu_{kj}^{x_{ij}} (1 - \mu_{kj})^{1 - x_{ij}}$

expected complete-data log-likelihood: $Q(\mu_{kj}) = \sum_i r_{ik} [x_{ij} \log \mu_{kj} + (1 - x_{ij}) \log (1 - \mu_{kj})]$

$$\frac{\partial Q}{\partial \mu_{kj}} = \sum_i r_{ik} \left[\frac{x_{ij}}{\mu_{kj}} - \frac{1 - x_{ij}}{1 - \mu_{kj}} \right]$$

$$\left(0 = \sum_i r_{ik} \left[\frac{x_{ij}}{\mu_{kj}} - \frac{1 - x_{ij}}{1 - \mu_{kj}} \right] \right) \mu_{kj} (1 - \mu_{kj})$$

$$0 = \sum_i r_{ik} [x_{ij} (1 - \mu_{kj}) - (1 - x_{ij}) \mu_{kj}]$$

$$0 = \sum_i r_{ik} (x_{ij} - \mu_{kj})$$

$$\sum_i r_{ik} x_{ij} = \mu_{kj} \sum_i r_{ik}$$

$$\boxed{\mu_{kj} = \frac{\sum_i r_{ik} x_{ij}}{\sum_i r_{ik}}}$$

This is the responsibility-weighted mean of the binary observations. □

Derive MAP update w/ Beta prior

we now assume $\mu_{kj} \sim \text{Beta}(a,b)$,

with density $p(\mu_{kj}) = \frac{1}{B(a,b)} \mu_{kj}^{a-1} (1-\mu_{kj})^{b-1}$

we want to show $\mu_{kj} = \frac{\sum_i r_{ik} x_{ij} + a - 1}{\sum_i r_{ik} + a + b - 2}$

For MAP estimation, we must maximize the expected complete-data log-likelihood plus the log prior:

$$Q_{MAP}(\mu_{kj}) = \sum_i r_{ik} [x_{ij} \log \mu_{kj} + (1-x_{ij}) \log(1-\mu_{kj})] + (a-1) \log \mu_{kj} + (b-1) \log(1-\mu_{kj}) + C$$

$$\frac{\partial Q_{MAP}}{\partial \mu_{kj}} = \frac{\sum_i r_{ik} x_{ij} + a - 1}{\mu_{kj}} - \frac{\sum_i r_{ik}(1-x_{ij}) + b - 1}{1-\mu_{kj}}$$

$$0 = \frac{\sum_i r_{ik} x_{ij} + a - 1}{\mu_{kj}} - \frac{\sum_i r_{ik}(1-x_{ij}) + b - 1}{1-\mu_{kj}}$$

$$\frac{\sum_i r_{ik} x_{ij} + a - 1}{\mu_{kj}} = \frac{\sum_i r_{ik}(1-x_{ij}) + b - 1}{1-\mu_{kj}}$$

Cross multiplying and collecting the μ_{kj} terms,

$$\sum_i r_{ik} x_{ij} + a - 1 = \mu_{kj} \left(\sum_i r_{ik} a + b - 2 \right)$$

Thus,

$$\boxed{\mu_{kj} = \frac{\sum_i r_{ik} x_{ij} + a - 1}{\sum_i r_{ik} a + b - 2}}$$

This is the interior MAP solution, valid when the posterior mode is interior (for example, when $a, b > 1$). Other parameter settings can produce boundary modes. □

Problem 2: Lasso Feature Selection

Derive the gradient

given objective: $J(x) = \|Ax - b\|_2^2 + \lambda \|x\|_1$.

since we know, $\|x\|_1 = \sum_{j=1}^{d} |x_j|$, $\frac{\partial}{\partial x_j}|x_j| = \text{sign}(x_j)$, $x_j \neq 0$,

we get: $\boxed{\nabla \|x\|_1 = \text{sign}(x)}$ where the sign function is applied elementwise, away from zero.

expanding squared residual term:

$$\|Ax-b\|_2^2 = (Ax-b)^T(Ax-b) = x^TA^TAx - 2b^TAx + b^Tb.$$

its gradient is: $\nabla_x \|Ax-b\|_2^2 = 2A^T(Ax-b)$

Combining the two terms:

$$\boxed{\nabla J(x) = 2A^T(Ax-b) + \lambda \, \text{sign}(x)}$$ for coordinates where $x_j \neq 0$.

At $x_j = 0$, the ℓ_1 term is not differentiable. Its subgradient is the interval $[-1, 1]$, so a subgradient-decent implementation can use zero as a valid choice.

the corresponding update is:

$$x^{(t+1)} = x^{(t)} - \eta_t \left[2A^T(Ax^{(t)} - b) + \lambda \, \text{sign}(x^{(t)})\right].$$